

Atomic Structure

Test · 24 marks total

Name: _____

Date: _____

Score: _____ / 24

Section A — Sub-atomic particles

Complete the table below.

Particle	Relative charge	Relative mass
Proton	_____	1
Neutron	0	_____
Electron	-1	_____

State where in the atom protons and neutrons are found: _____

[4 marks]

Section B — Counting particles

B1. Aluminium has atomic number 13 and mass number 27. Calculate the number of neutrons, and state the number of electrons.

[2 marks]

B2. An atom has atomic number 20 and mass number 40. Calculate the number of neutrons, and state the number of electrons.

[2 marks]

Section C — Isotopes

C1. Define the term “isotope”.

[2 marks]

C2. Carbon-12 has 6 protons and 6 neutrons. Carbon-14 has 6 protons and 8 neutrons. Explain why these are both still carbon, but have different masses.

[2 marks]

Section D — Electronic structure

Give the electronic structure of an atom with each atomic number below.

D1. Atomic number 7 (nitrogen). [2 marks]

D2. Atomic number 14 (silicon). [2 marks]

D3. Atomic number 19 (potassium). [2 marks]

Section E — Development of the atomic model

E1. Describe the plum pudding model of the atom.

[2 marks]

E2. Describe the results of the alpha particle scattering experiment, and explain what conclusions were drawn about atomic structure as a result.

[4 marks]